Zewail City of Science, Technology and Innovation

University of Science and Technology

School of Computational Sciences and Artificial Intelligence

GRADUATION PROJECT

MARKET VISIBILITY REPORT

QuranGPT: A RAG-Based Retrieval and Question-Answering System for the Quran and Tafseer

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1. Executive Summary

Quranic study often requires locating relevant verses and interpreting them through established Tafsir works. This is difficult when users ask questions in contemporary or everyday Arabic that differs from classical source wording. Keyword search can miss semantically related passages, while general-purpose language models may produce fluent but unsupported interpretations or inaccurate citations. QuranGPT proposes an Arabic-first retrieval-augmented generation (RAG) web system that grounds answers in structured Quranic and Tafsir evidence from Quranpedia data dumps. The project will normalize and validate Quranic verses and selected Tafsir passages, preserve source and verse metadata, and index the material for semantic retrieval and structured verse lookup. Retrieved passages will be supplied to a language model to generate readable answers accompanied by traceable surah, ayah, and Tafsir references. The intended users are Quran learners, students, and readers seeking accessible explanations they can verify against source texts. Related work includes Quran-focused question answering, shared-task passage retrieval, Tafsir-aware RAG, and recent grounded Islamic QA benchmarks. Existing platforms provide valuable Quran search, Tafsir, linguistic resources, or recitation tools, but they do not all target the same integrated Arabic question-to-Tafsir workflow. QuranGPT will evaluate retrieval relevance, citation validity, answer support, and latency rather than claim accuracy in advance. The first-semester goal is a working cited-QA prototype and baseline evaluation; second-semester work may add hybrid retrieval, reranking, and citation verification. It is a study aid, not an authority for independent religious rulings.

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2. Problem Statement & Motivation

2.1 Core Problem Definition

Users need a reliable way to retrieve relevant Quranic and Tafsir evidence and receive understandable Arabic answers with verifiable references, without unsupported generation or misattribution of source passages.

2.2 Problem Context & Real-World Impact

Quranic explanations are distributed across verses and multiple Tafsir books, whose terminology, length, and structure vary. A learner may phrase a question in Modern Standard Arabic or everyday language that does not appear verbatim in classical texts. Keyword-based search may consequently fail to retrieve passages that are conceptually relevant. At the other extreme, a general language model can answer fluently while inventing quotations, citing the wrong ayah, or blending explanations from different sources. The affected stakeholders include secondary and university students, independent learners, educators, and readers who want to check an explanation against its source. Poor retrieval or unclear attribution increases the effort required to verify information and can undermine trust in digital study tools. The proposal does not contain measured user time-loss, error-rate, or economic-impact figures; these impacts are therefore qualitative and will be investigated through project evaluation.

2.3 Motivation & Urgency

The project is timely because machine-readable Quran and Tafsir resources are available through Quranpedia's versioned JSON dumps, while Arabic NLP embeddings and RAG pipelines make semantic question-to-passage retrieval practical. Quranpedia documents separate Quran, ayah-service, Tafsir-book, and metadata files, enabling reproducible ingestion rather than scraping a website [6]. At the same time, recent Islamic QA research explicitly studies hallucination, abstention, and evidence grounding [5]. These developments create an opportunity to build a focused educational tool while treating religious source fidelity as a core engineering requirement. The urgency is not a claim of measured market demand; it is the combination of an accessible data source, feasible current methods, and the need for transparent verification.

3. Proposed Solution & Program Focus

3.1 Solution Overview & System Workflow

The proposed system follows an offline data-preparation stage and an online question-answering stage. During ingestion, the team will obtain the Quranpedia JSON dumps, select the Quran text and at least two Tafsir works, parse and normalize records, validate required fields, and retain provenance such as book, source, surah, ayah or ayah range, and passage identifier. The validated records will be stored and indexed for semantic retrieval, with structured verse lookup available for exact references. At query time, a user submits an Arabic question through the web interface. The retriever ranks relevant verse and Tafsir passages; the selected evidence is passed to an LLM with instructions to answer only from the supplied context and identify its sources. The interface presents the answer alongside clickable source references and original passages. If the retrieved evidence is insufficient, the intended safe behavior is to communicate uncertainty rather than fabricate an answer. Hybrid lexical-semantic retrieval, reranking, and claim-level citation checks are planned enhancements subject to time and evaluation results.

3.2 Key System Features

Feature 1 – Arabic Question Answering: Users can submit natural-language questions in Arabic and receive a clear response generated from retrieved Quranic and Tafsir evidence.

Feature 2 – Evidence Retrieval and Traceable Citations: The system retrieves relevant verses and Tafsir passages and displays surah, ayah, book, and passage references so users can inspect the original sources.

Feature 3 – Source-Aware Study Interface: A web interface presents the generated explanation together with supporting source text; multi-Tafsir comparison and stronger citation verification are planned extensions.

3.3 Track-Specific Technical Focus

The project is in the Data Science and Artificial Intelligence (DSAI) track.

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DSAI pipeline: (1) Data collection from Quranpedia's versioned JSON dumps; (2) schema inspection and selection of the Quran text plus at least two Tafsir works; (3) Arabic text normalization that preserves the canonical source text and records any normalized search representation separately; (4) integrity checks for missing text, duplicate identifiers, invalid surah/ayah references, and provenance fields; (5) passage segmentation and metadata-aware indexing; (6) semantic embedding retrieval with structured verse lookup; (7) context assembly and LLM-based grounded answer generation; and (8) evaluation. The planned evaluation set contains at least 100 representative questions. Retrieval will be measured using Recall@5 and Mean Reciprocal Rank (MRR); citation checks will measure reference validity, an expert-reviewed subset will assess answer support/faithfulness, and system performance will include response latency. Semester 2 will compare the baseline retriever with at least one enhanced approach, such as hybrid retrieval or reranking. No specific embedding model, vector database, LLM provider, or framework is fixed in the proposal; these will be selected and documented during implementation. The team will not train an LLM from scratch.

4. Target Audience & User Persona Analysis

4.1 Primary User Persona

Primary persona: an Arabic-speaking university or secondary-school student, independent learner, or beginning Quran student who is comfortable using a web application but may not have specialist training in classical Arabic or Tafsir. The user may know the topic or question they want to understand but not the exact surah, ayah, or terminology used in reference books. A secondary persona is an educator or study-group facilitator who wants to locate source passages efficiently and direct learners to the original references.

4.2 User Characteristics, Goals, and Usage Context

Users are likely to access the system from a laptop or mobile browser during individual study, coursework, or guided learning. Their goals include finding relevant verses, understanding how selected Tafsir works explain them, comparing source passages, and verifying where an answer came from. Common pain points are unfamiliar classical vocabulary, long source texts, inconsistent terminology, and uncertainty about whether an AI-generated answer is actually supported. The system should therefore support Arabic questions, readable answers, visible source attribution, and direct inspection of the retrieved text. The first release focuses on Arabic text input; speech input is optional future work.

4.3 User Need & Value Proposition

The value proposition is a shorter path from a natural-language question to relevant, inspectable Quranic and Tafsir passages. Rather than asking users to trust a free-form generated answer, QuranGPT makes its evidence visible and preserves the identity of the source book and verse reference. This can reduce manual navigation and make verification easier, although no time-saving percentage or accuracy gain is claimed before user testing. For educators, the same provenance can support discussion and source checking; for learners, it provides an accessible starting point for further study while leaving scholarly interpretation to qualified people.

5. Literature Review & Background Research

5.1 Background & Key Concepts

Retrieval-augmented generation (RAG) combines an information-retrieval stage with a generative language model. Instead of relying only on the model's learned parameters, the system retrieves external passages for a query and supplies them as context for answer generation. For QuranGPT, the retriever operates over structured Quranic verses and selected Tafsir passages. Semantic retrieval uses text embeddings to identify passages with related meaning even when query wording differs; structured lookup handles exact surah/ayah references. A later hybrid system may combine semantic similarity with lexical matching, and a reranker may reorder the initial candidate set. Arabic preprocessing must be handled carefully: normalization can help matching across orthographic variants, but the original Quran and Tafsir text must remain unchanged for display and citation. Provenance metadata links every passage to its book and verse/range. Evaluation separates retrieval quality (Recall@5, MRR) from citation validity, answer support/faithfulness, and latency. A responsible system should disclose insufficient evidence and avoid presenting generated text as a religious ruling.

5.2 Summary of Related Work

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
[1]	Abdelnasser et al., 2014	Al-Bayan: specialized Arabic Quran question-answering system; retrieval and QA pipeline.	Holy Quran; Arabic religious QA context.	Demonstrated an early practical Quran-focused Arabic QA system and highlighted the need for domain-specific accuracy.	Pre-LLM approach; does not evaluate modern RAG generation or claim-level citation faithfulness.
[2]	Malhas, Mansour & Elsayed, 2023	Shared task separating Quranic passage retrieval (PR) and machine-reading comprehension (MRC).	Qur'an QA 2023 benchmark; MSA questions and Quranic passages; multiple participating teams.	Established reusable tasks and evaluation settings for ranking answer-bearing passages and extracting answers.	Task format does not by itself assess long-form Tafsir synthesis, source attribution across books, or generated-answer faithfulness.
[3]	Zekiye & Amroush, 2023	Al-Jawaab at Qur'an QA 2023; dense embeddings combined with GPT-4 for question answering.	Qur'an QA 2023 task setting and Quranic passages.	Illustrates the feasibility of combining embedding-based retrieval with an LLM for Quran QA.	The proposal's preliminary citation should be checked against the final bibliography; system-level source verification and multi-Tafsir evaluation remain to be assessed.
[4]	Al-Azani et al., 2025	OntologyRAG-Q; ontology-aware, verse-level Tafsir chunking with retrieval and LLM benchmarking.	15 structured Tafsir books; approximately 4,200 QA pairs; annotated Tafsir ontology.	Provides structured resources and benchmarks for Tafsir retrieval and RAG evaluation.	Multi-hop/contextual reasoning and dependable support for unrestricted user questions remain challenging.

Ref.	Author(s), Year	Method / Approach	Dataset / Context	Key Results	Limitations
[5]	Bhatia et al., 2026	Grounded Islamic QA suite and agentic RAG with iterative evidence seeking and answer revision.	IslamicFaithQA: 3,810 bilingual Arabic/English items; ~6k verse retrieval corpus; additional grounded training/preference resources.	Reports retrieval-related correctness gains and stronger performance from agentic evidence seeking; includes explicit hallucination/abstention evaluation.	Broader Islamic QA scope and agentic complexity differ from a lightweight Quranpedia-based student prototype; results do not guarantee performance on this project's corpus.

5.3 Critical Synthesis & Research Gap

The literature shows a progression from specialized Arabic Quran QA [1], to shared evaluation of passage retrieval and reading comprehension [2], to embedding-assisted generation [3], structured Tafsir resources and ontology-aware RAG [4], and broader grounded Islamic QA benchmarks that explicitly test hallucination and abstention [5]. Across these works, retrieval is central: the system must find the correct source material before generation can be trusted. The 2023 shared task is particularly relevant because it separates passage retrieval from reading comprehension, making retrieval errors measurable [2]. OntologyRAG-Q contributes a large structured Tafsir resource and verse-aware chunking [4], while IslamicFaithQA broadens evaluation beyond conventional MCQ/MRC settings to free-form answers and abstention [5]. The proposal-cited Al-Jawaab work explores embeddings with GPT-4 [3]. These works provide strong foundations, but they do not remove the need to validate a specific deployed corpus, preserve passage-level provenance, and evaluate the end-to-end answer/citation behavior on the chosen data. QuranGPT's intended contribution is an integrated, reproducible Quranpedia-based educational prototype with visible original evidence and a measured baseline. It does not claim to invent RAG, Arabic semantic search, or Quranic QA.

6. Market Visibility: Existing Solutions & Competitive Analysis

6.1 Review of Similar Systems & Existing Solutions

Solution / Provider	Type	Description & Core Functionality	Target Users	Availability / Pricing
Quran.com / Quran Foundation	Nonprofit platform + developer APIs	Quran text, translations, Tafsir, audio, search, bookmarks/notes and developer APIs; documented content and search endpoints.	General Quran readers, learners, developers and researchers.	Website and mobile apps; public content access and developer APIs documented. Verify current API credentials/terms before integration.
Quranic Arabic Corpus	Academic / research resource	Annotated Quranic Arabic resource with morphology, syntax/treebank, semantic ontology, and keyword/stem/lemma/root search.	Arabic/NLP researchers, students of Quranic language and advanced learners.	Public research website; search and linguistic data available online.

Solution / Provider	Type	Description & Core Functionality	Target Users	Availability / Pricing
Tarteel	Commercial/ freemium AI application	AI-assisted Quran recitation, voice search, memorization mistake detection, listening, goals and progress tracking; Tafsir reading tools are also listed in its help center.	Quran readers, reciters and memorization students.	iOS/Android; free features plus paid Premium subscription for selected AI/memorization features.

6.2 Feature Comparison Matrix

Evaluation Feature	Your Solution	Competitor A: Quran.com	Competitor B: Quranic Arabic Corpus	Competitor C: Tarteel
Core Technology	Planned Arabic RAG: semantic retrieval + structured verse lookup + LLM generation.	Quran content/search APIs and web/mobile content platform.	Linguistic annotation, morphology, syntax, ontology and search.	Speech/recitation AI, voice search and memorization support.
Key Features	Arabic natural-language QA; retrieved Tafsir evidence; clickable source citations; source-aware answer display.	Quran text, translations, Tafsir, audio, keyword/topic search, bookmarks and notes.	Word-level morphology, grammar, syntactic trees, ontology and lexical search.	Recitation follow-along, voice search, mistake detection, memorization goals; Tafsir reading tools.
Performance / Accuracy	Not measured yet. Planned Recall@5, MRR, citation validity, expert-reviewed answer support and latency.	No directly comparable end-to-end cited Arabic Tafsir QA metric established in reviewed product documentation.	Designed for linguistic lookup; no directly comparable generative QA metric identified.	Vendor describes recognition/memorization capabilities; not directly comparable to Tafsir QA metrics.
User Experience	Planned web interface focused on asking an Arabic question and inspecting retrieved evidence.	Mature reading and study experience across web/apps.	Research-oriented interface with specialized linguistic search.	Mobile-first recitation and memorization workflow; web/computer support documented.
Integration Capability	Planned internal ingestion/indexing; future integration choices not fixed.	Documented content, search and user APIs for developers.	Public research resource; structured linguistic information can support analysis.	Primarily an end-user application; public product pages reviewed do not establish an equivalent open Tafsir-QA API.
Platform & Availability	Planned browser-based prototype; deployment details not finalized.	Web and mobile apps; developer documentation available.	Public website and online corpus.	iOS and Android; computer access guidance available in support center.
Cost Structure	No project pricing model; academic prototype and model/hosting costs to be assessed.	Quran.com describes its public service as free and nonprofit; API terms/credentials apply.	Publicly accessible research resource.	Freemium; Premium subscription for selected AI features.

6.3 Strengths & Weaknesses Analysis

Competitor	Strengths	Weaknesses / Limitations	Implication for Your Solution
Quran.com	Broad Quran content, translations, Tafsir, audio, search, user study features, and documented APIs.	Its main product is a comprehensive Quran reading/study platform; the reviewed documentation does not establish the exact controlled Quranpedia-based Arabic question-to-multi-Tafsir RAG workflow proposed here.	Use its clear source navigation and API documentation as design references; differentiate through focused question-led retrieval and explicit evaluation, not by claiming broader content coverage.
Quranic Arabic Corpus	Detailed morphology, syntax, semantic ontology, and powerful linguistic search useful for Arabic research.	Specialized linguistic analysis can be demanding for casual learners; it is not presented as a conversational Tafsir answer-generation product.	Preserve linguistic/source metadata where useful, but prioritize a simple natural-language interface and explanations grounded in selected Tafsir passages.
Tarteel	Strong recitation-centered experience, voice search, memorization support, progress tools, and accessible mobile apps.	Its central workflow is recitation and memorization rather than evidence-grounded explanatory QA across Tafsir books.	Adopt the principle of task-focused interaction; keep speech input optional and do not expand scope into recitation assessment.

6.4 Gap Analysis & Key Differentiators

#	Identified Gap	Evidence (Lit. / Competitor Ref.)	How Your Solution Addresses It
1	Natural-language Arabic question-to-Tafsir retrieval is not the central workflow of all existing Quran study tools.	[1]–[4]; Quran.com [7] and Tarteel [9] emphasize broad reading/search or recitation workflows.	Build a focused Arabic QA flow that retrieves relevant verses and selected Tafsir passages for a user's question.
2	A fluent generated explanation can obscure whether each reference and claim is supported by the cited source.	[5] identifies free-form hallucination and abstention as important evaluation dimensions; [4] addresses Tafsir-specific RAG resources.	Display original retrieved passages and source metadata; evaluate reference validity and expert-reviewed answer support; abstain when evidence is insufficient where feasible.
3	Retrieval performance and answer faithfulness need to be measured on the project's actual selected corpus, not assumed from another benchmark.	[2], [4], [5]; the competitor product documentation reviewed does not provide directly comparable Recall@5/MRR and citation-support results.	Create at least 100 representative questions and report Recall@5, MRR, citation validity, answer support on a reviewed subset, and latency; compare baseline with an enhanced retriever.

QuranGPT's differentiator is its planned combination of (a) Quranpedia-based local ingestion and validation, (b) Arabic question answering over Quranic text and a deliberately bounded selection of at least two Tafsir works, (c) visible verse- and book-level provenance, and (d) an evaluation plan that measures retrieval, citation validity, answer support, and latency. Quran.com already provides extensive Quran content, Tafsir, search, and developer APIs [7]; Quranic Arabic Corpus offers deep linguistic annotation and search [8]; and Tarteel focuses on recitation, voice search, and memorization workflows [9]. QuranGPT is not positioned as a replacement for these resources. It focuses on the narrower workflow of asking a question in Arabic and receiving a source-grounded explanation from a controlled Quranpedia-derived corpus. Whether it offers better retrieval or usability remains an empirical question for the planned evaluation.

7. Project Roadmap & Deliverables

The roadmap below follows the proposal's two-semester plan. Exact week-level dates may be adjusted to the

department's GP calendar.

Project Timeline & Work Breakdown

#	Phase	Key Activities	Start / End	Milestone
1	Requirements, proposal and literature review	Finalize problem, scope, user stories, risks and initial literature/market matrix; inspect Quranpedia dump structure and usage terms.	Oct–Nov 2026	Approved proposal and requirements/SRS baseline (align with department Week 10 checkpoint).
2	Data inspection and initial design / proof of concept	Inspect JSON schemas and metadata; choose at least two Tafsir works; define normalized record schema, provenance fields, preprocessing and index design; test a small retrieval proof of concept.	Nov–Dec 2026	Documented data schema, source-selection rationale and initial PoC (target Week 12).
3	Semester 1 core MVP	Normalize and validate Quran/Tafsir data; build semantic index and structured lookup; implement retrieval, grounded LLM response generation, backend and web interface with clickable citations.	Nov 2026–Jan 2027	End-to-end Arabic cited-QA prototype and draft report (target Week 13).
4	Semester 1 evaluation and Part 1 defense	Prepare at least 100 representative questions; measure Recall@5/MRR, citation validity, expert-reviewed answer support on a subset, and latency; document limitations and demo.	Dec 2026–Jan 2027	Baseline evaluation report and Part 1 defense package (target Week 15).
5	Semester 2 improvements and final delivery	Compare baseline with hybrid retrieval and/or reranking; investigate claim-level citation verification; optional speech input or feedback-driven experiments only if resources permit; finalize testing, report and demonstration.	Feb–Jun 2027	Comparative results, final system/demo, documentation and final report.

8. Risk Management & Professional Ethics

Identified Risk	Likelihood / Impact	Mitigation Strategy
Inconsistent or incomplete Quranpedia records, metadata, or verse mappings	Medium / High	Inspect schemas before ingestion; validate required fields and surah/ayah ranges; retain source IDs; log rejected records; spot-check against original source pages.
Arabic normalization harms Quranic text fidelity or retrieval	Medium / High	Keep immutable original text for display; store normalized search text separately; test diacritics/orthographic variants; manually inspect representative cases.
Retriever returns irrelevant Tafsir passages or misses evidence	Medium / High	Create a representative evaluation set; establish a baseline; inspect failure cases; compare hybrid retrieval/reranking in Semester 2; avoid unsupported answers when evidence is weak.
LLM hallucinates interpretations, quotations, or citations	Medium / High	Constrain generation to retrieved context; render citations from retrieved metadata rather than free-form model text; verify reference IDs; include uncertainty/abstention behavior; review a sample with knowledgeable reviewers.

Identified Risk	Likelihood / Impact	Mitigation Strategy
Tafsir data usage, attribution, or licensing conditions are misunderstood	Low / High	Read the Quranpedia usage policy and per-file license before redistribution or deployment; retain attribution/version information where required; do not assume every included book has identical rights.
LLM/API cost, rate limits, or service availability affect prototype	Medium / Medium	Estimate query volume and cost early; cache where appropriate; use a modular provider interface; document fallback behavior and keep the prototype's scope bounded.
Schedule overrun due to optional features (voice, RL, multi-hop/agentic behavior)	Medium / High	Prioritize the Semester 1 end-to-end MVP and evaluation; treat speech and feedback/RL as optional; only add enhancements after baseline milestones are met.
Insufficient expert review or biased/incomplete evaluation questions	Medium / High	Document question sources and annotation criteria; use multiple reviewers where possible; report subset size and limitations; do not generalize beyond the tested set.

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